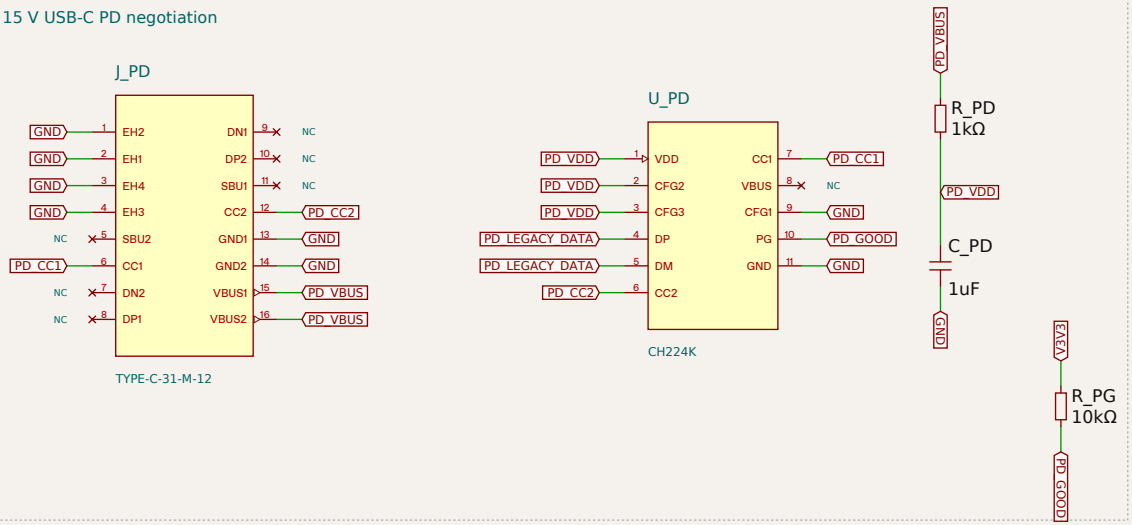
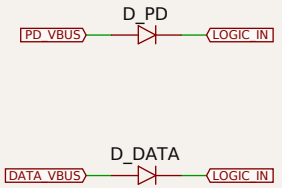
NEMA14 RP2040 | USB-C PD & Logic Power

15 V USB-C PD negotiation



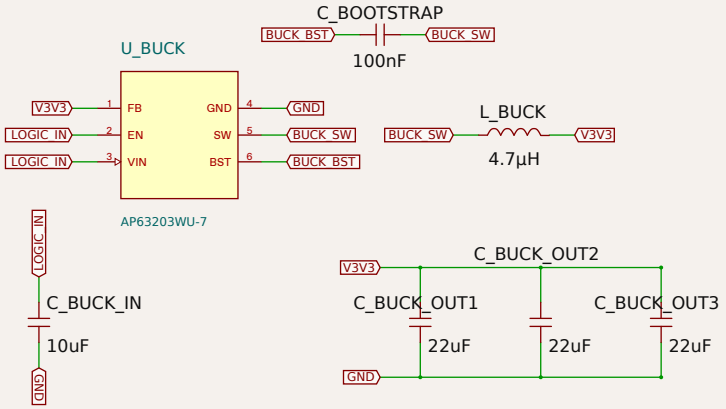
U_PD / CH224K requests 15 V (CFG1=0, CFG2=CFG3=1).
Open-drain PG is active low. DP/DM short: PD-only mode.
R_PD + C_PD supply/filter the negotiation IC.

USB / PD diode OR



Schottky diodes accept either supply and block crossfeed.

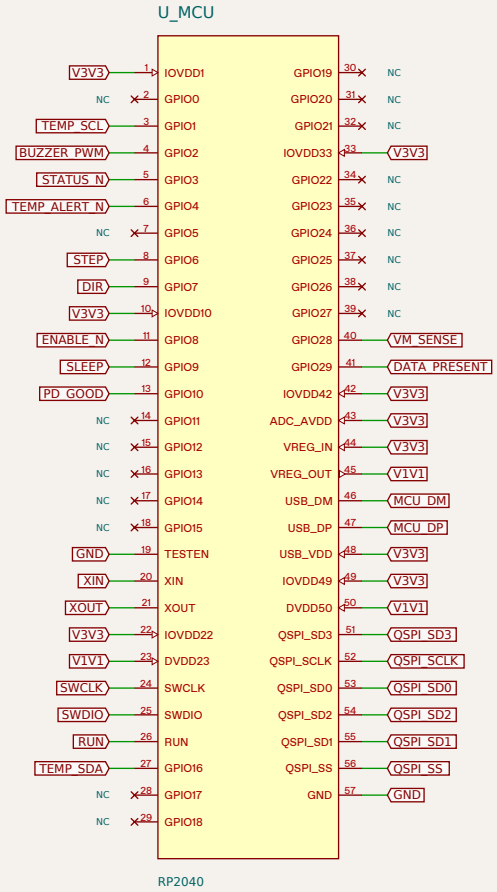
3.3 V buck supply



U_BUCK / AP63203 converts LOGIC_IN to 3.3 V.
4.7 uH L_BUCK and output capacitors filter the switching node.
C_BOOTSTRAP supports the high-side gate driver.

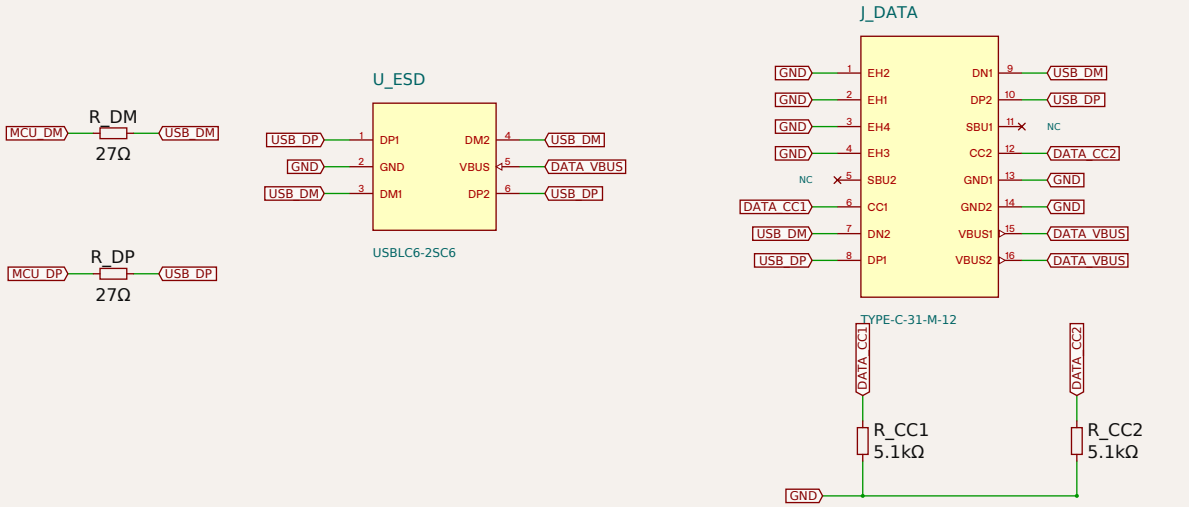
NEMA14 RP2040 | RP2040 & USB Control

Application MCU / physical pin map



U_MCU / RP2040 generates STEP/DIR and reads power status.
Internal regulator: 1.1 V core. Flash/clock/bypass: next sheet.

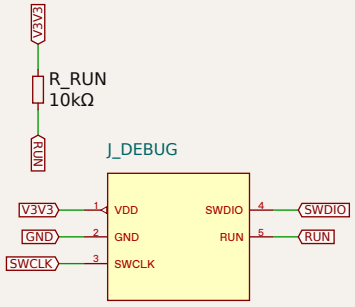
Computer USB-C & ESD protection



U_ESD / USBLC6 protects D+/D-. R_CC1/2 identify a USB sink.

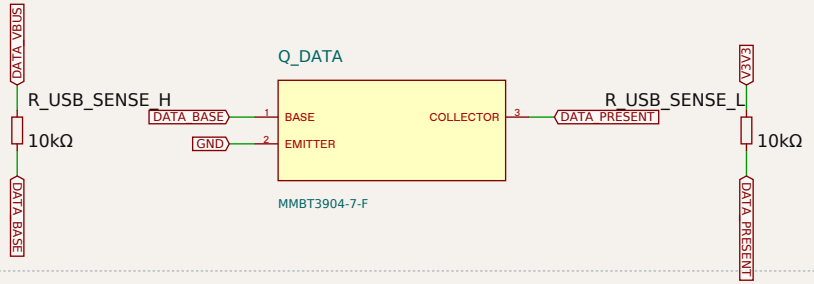
Programming & recovery

R_RUN holds reset high; SWD / RUN pads allow debug.
USB ROM boot: BOOT pads on next sheet.



PCB recovery / debug pads

Host VBUS detection

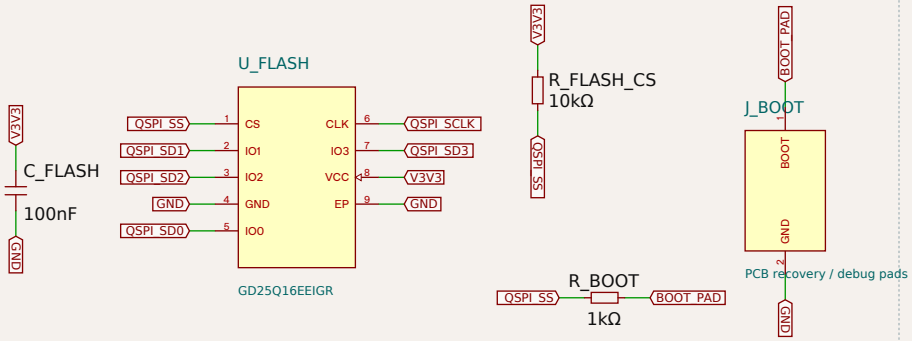


Q_DATA / MMBT3904 senses host 5 V without a direct GPIO feed.
DATA_PRESENT LOW means host attached.

A4 landscape | 2 / 5 | Named nets connect across sheets | NC = intentionally unused pin

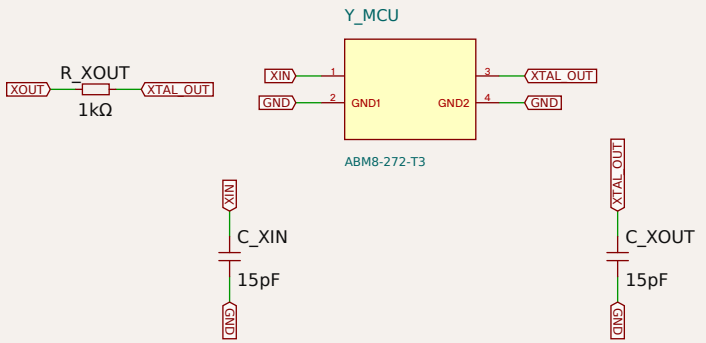
NEMA14 RP2040 | Flash, Clock & MCU Supply

External QSPI firmware & ROM recovery



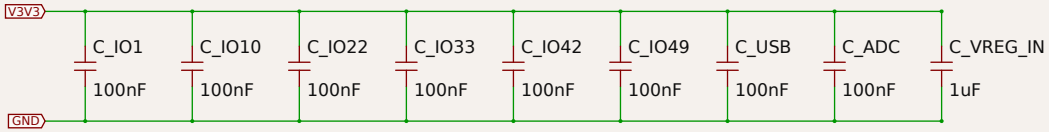
U_FLASH / GD25Q16 provides 2 MB firmware storage.
RP2040 has no internal flash. BOOT shorts QSPI_SS via R_BOOT to ground during reset, entering the ROM USB bootloader.

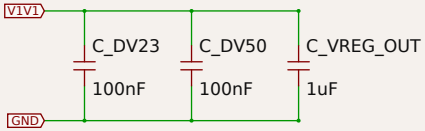
12 MHz USB clock reference



Y_MCU / ABM8 is the USB clock reference.
Load capacitors support oscillation; R_XOUT limits crystal drive.

MCU power pins / local bypass



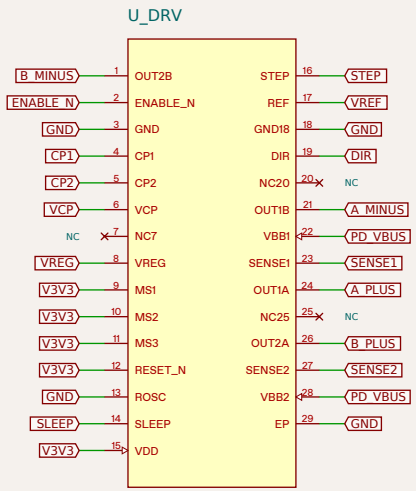


Each capacitor is a distinct fitted part.
IO pin numbers appear in references.
VREG_OUT feeds V1V1; no external 1.1 V source.

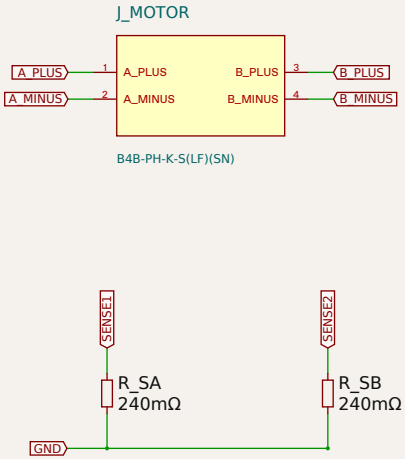
A4 landscape | 3 / 5 | Named nets connect across sheets | NC = intentionally unused pin

NEMA14 RP2040 | Motor Driver & Voltage Monitor

A4988 / bipolar winding driver

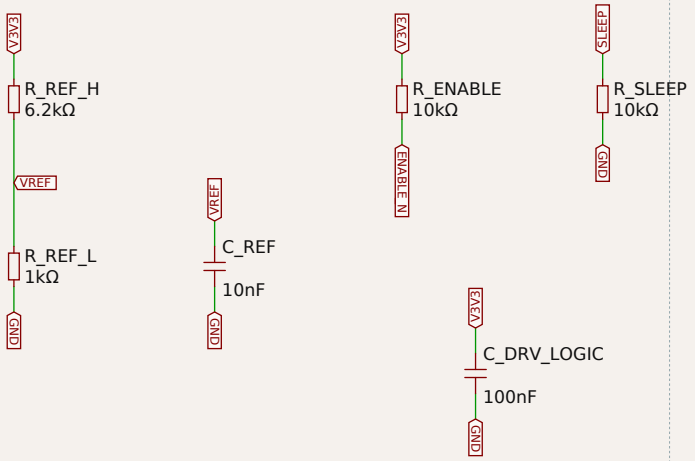


Keyed motor output & current shunts



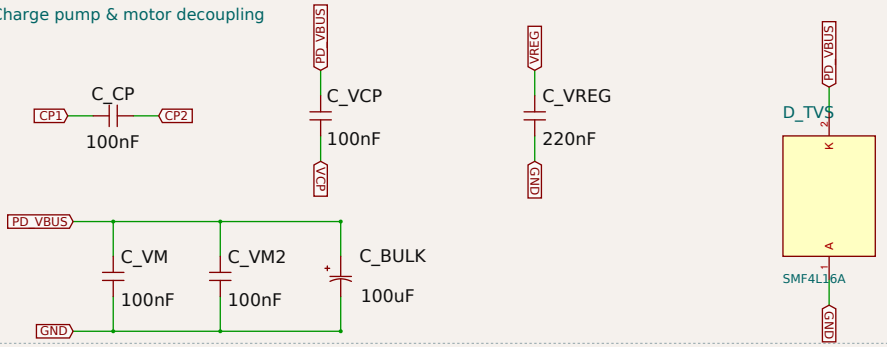
1 Black A+ 2 Green A-
3 Red B+ 4 Blue B-

Current reference & safe defaults



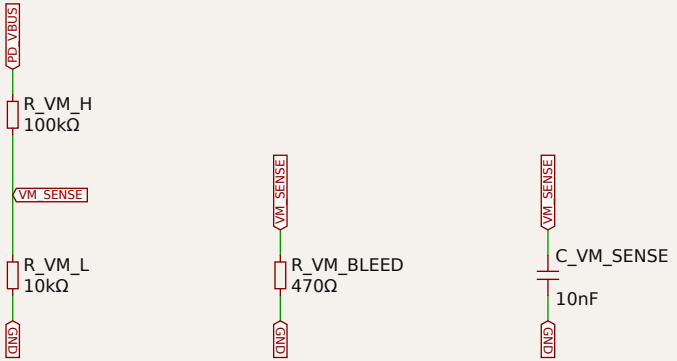
6.2k / 1k VREF and 0.24 ohm shunts: 0.239 A nominal.
ENABLE pull-up / SLEEP pull-down keep the driver off.

Charge pump & motor decoupling



U_DRV regulates both windings at fixed 1/16 microsteps.
C_CP / C_VCP / C_VREG support the internal driver rails.
Bulk capacitance and TVS limit transients; regeneration needs bench tests.

Motor-voltage monitor / power-off isolation

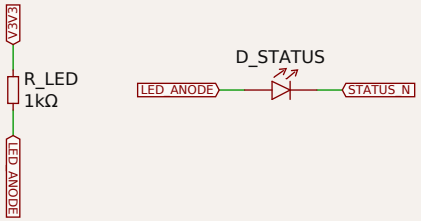


Passive 100k / (10k || 470R) divider: VM / 223.77.
Input stays below 0.3 V even when MCU power is absent.

A4 landscape | 4 / 5 | Named nets connect across sheets | NC = intentionally unused pin

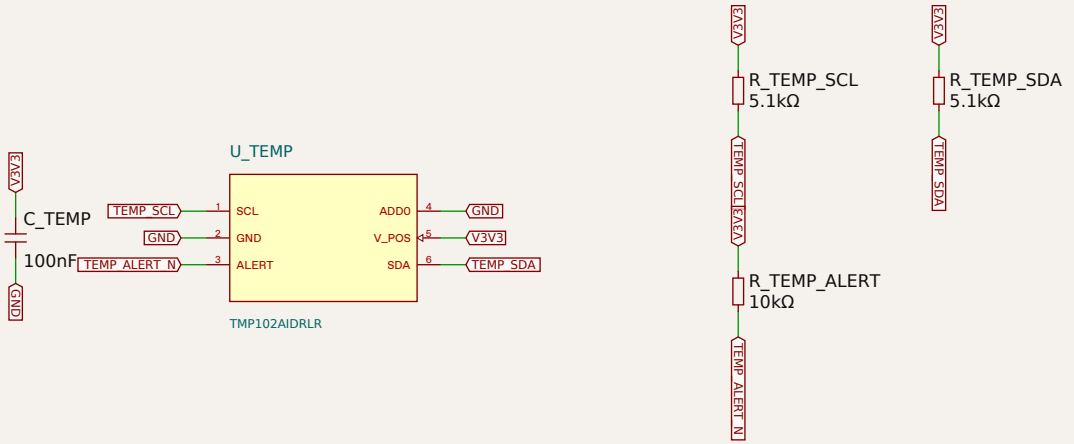
NEMA14 RP2040 | Status, Temperature & Audible Alarm

Low-current green status LED



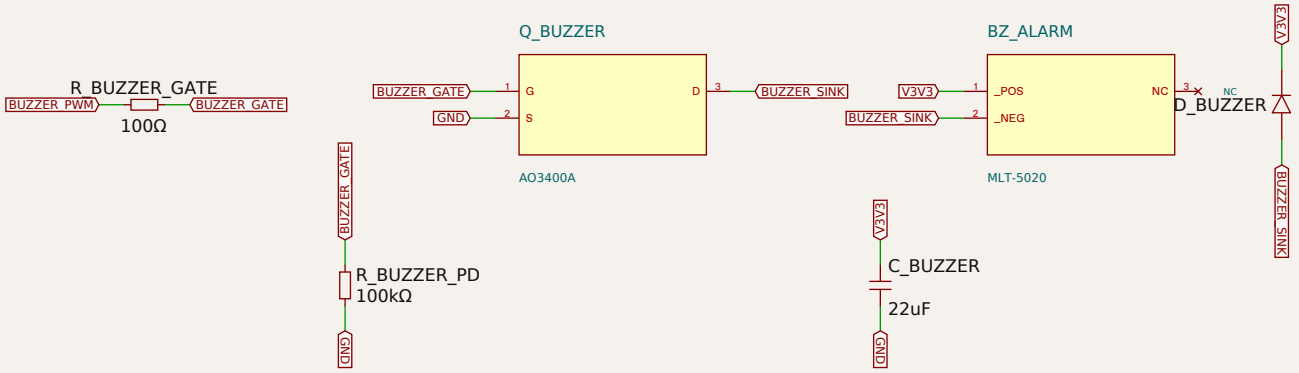
D_STATUS + 1k R_LED: active-low MCU status indicator.
Power-up input/high = off; low = on. Verify visibility.

Local board-temperature telemetry



U_TEMP / TMP102: I2C address 0x48 (ADD0 grounded).
5.1k SCL/SDA pull-ups; 10k ALERT pull-up; 100nF bypass.
Reports local PCB temperature, not measured motor/winding temperature.

PWM sounder / default-off transistor drive



MLT-5020 sounder: external PWM near 4 kHz; default OFF.
AO3400A low-side switch; 100 ohm gate + 100k pull-down.
1N5819WS flyback clamp + 22uF bypass. Acoustic/current tests pending.

A4 landscape | 5 / 5 | Named nets connect across sheets | NC = intentionally unused pin